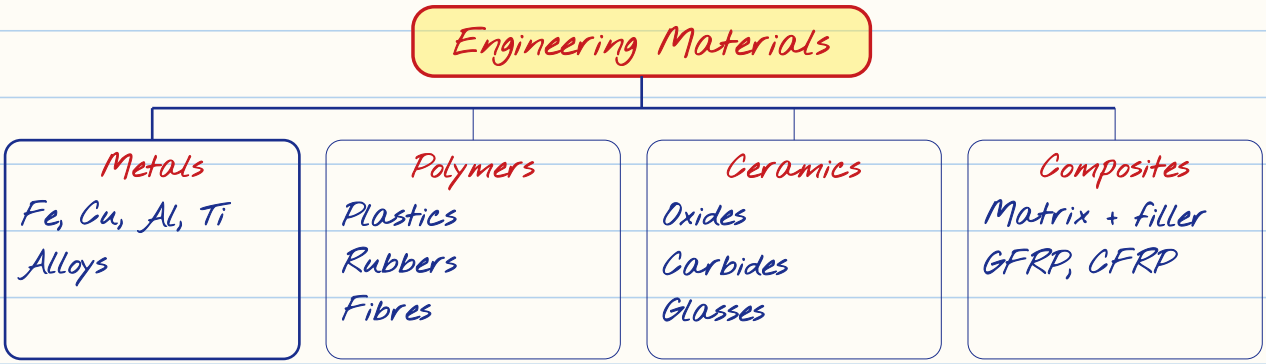
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UNIT-I : INTRODUCTION

IMPORTANT: Syllabus: Characteristics of metals, polymers and ceramics. Introduction to smart materials. Classification of smart materials. Components of a smart system. Applications of smart materials.

1.1 Engineering Materials - Overview

- Materials are the foundation of technology ; the properties of a material decide where it can be used.
- Engineering materials are mainly classified into four families:



- Properties depend on atomic bonding , crystal structure and microstructure.
  - Metals → metallic bond; Polymers → covalent (chains) + weak secondary bonds; Ceramics → ionic / covalent bond.
- Conventional materials are passive : they cannot change their properties by themselves when the environment changes.
- Smart materials are active : they change shape, stiffness, colour, conductivity, etc. in response to a stimulus.

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1.2 Characteristics of Metals

Definition: Metals are crystalline materials made of atoms held by the **metallic bond** – positive ion cores surrounded by a "sea" of free (delocalised) electrons.

Important characteristics

- **High electrical conductivity** and **high thermal conductivity** because of free electrons.
- **Ductile** (can be drawn into wires) and **malleable** (can be hammered into sheets) – because atomic planes slip easily over each other.
- **High strength** and **toughness**; good fatigue and impact resistance.
- Shiny, opaque and have a metallic **lustre** (free electrons reflect light).
- High melting point (mostly) and high density; most are good heat conductors.
- Can be strengthened by **alloying**, cold working, heat treatment and grain refinement.
- Can be recycled easily; machinable, weldable, castable and forgeable.
- Generally **crystalline** with structures BCC (Fe, Cr), FCC (Cu, Al, Ni) and HCP (Ti, Zn, Mg).

Types of metals

- **Ferrous** metals (iron-based): cast iron, mild steel, tool steel, stainless steel.
- **Non-ferrous** metals: aluminium, copper, titanium, magnesium, zinc, nickel and their alloys (brass, bronze).
- **Alloys** = mixture of a metal with other elements to improve properties, e.g. steel (Fe + C), brass (Cu + Zn).

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Limitations

- Corrosion (rusting), heavy weight, high thermal expansion, electrical short-circuit risk, and fatigue failure under cyclic loads.

Applications: Wires and cables (Cu), aircraft body (Al, Ti), bridges and buildings (steel), engines, tools, electronics (Au, Ag contacts), heat exchangers.

1.3 Characteristics of Polymers

Definition: Polymers are **macromolecules** made of a large number of small repeating units (**monomers**) joined by covalent bonds. Poly = many, mer = part.

- Polymerisation: ethylene (monomer) → polyethylene (polymer).
- Held together in bulk by weak **secondary bonds** (van der Waals, hydrogen bonds) between long chains.

Important characteristics

- **Low density** (light weight), low cost and easy to mould into complex shapes.
- **Electrical and thermal insulators** (no free electrons) - used for cable insulation.
- Good **corrosion and chemical resistance** ; do not rust.
- Low strength, low stiffness and low melting / softening point compared to metals.
- **Viscoelastic** behaviour: partly viscous + partly elastic, properties depend on time, temperature and rate of loading.
- Can be transparent, coloured, flexible or rigid by choosing the structure.
- Degrade with UV light, heat and oxygen (ageing); disposal / recycling is an environmental problem.

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Classification of polymers

Type	Structure	Behaviour on heating	Examples
Thermoplastics	Linear / branched chains	Soften on heating, reshaped again	PE, PVC, PS, nylon
Thermosets	Highly cross-linked 3-D network	Do not soften, char and decompose	Epoxy, bakelite, polyester
Elastomers	Lightly cross-linked chains	Large elastic stretch (up to 500%)	Natural rubber, silicone, neoprene

Note: Degree of polymerisation (DP) = molecular weight of polymer / molecular weight of one mer. Higher DP → higher strength and melting point.

Applications: Packaging, pipes, cable insulation, tyres, fibres, adhesives, coatings, biomedical implants, sensors (PVDF, conducting polymers).

1.4 Characteristics of Ceramics

Definition: Ceramics are inorganic, non-metallic solids (compounds of metals with oxygen, carbon, nitrogen) formed by heating at high temperature (sintering) and then cooling.

- Bonding is ionic and/or covalent - very strong, directional bonds, so atoms cannot slide easily.

Important characteristics

- Very hard and wear resistant (alumina, silicon carbide, diamond-like).
- High melting point and excellent high-temperature stability (refractories).
- High compressive strength but brittle - low tensile strength, low fracture toughness, fail suddenly.

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- Electrical and thermal insulators (some are semiconductors or even superconductors, eg. YBCO).
- Chemically inert, good corrosion resistance in acids and alkalies.
- Low thermal expansion and low density compared to metals; poor thermal shock resistance.
- Some ceramics are functional : piezoelectric (PZT, $BaTiO_3$), ferroelectric, magnetic (ferrites), dielectric.

Types

- Traditional : clay, bricks, tiles, porcelain, cement, glass.
- Advanced / engineering : alumina (Al_2O_3), zirconia (ZrO_2), silicon carbide (SiC), silicon nitride (Si_3N_4).
- Smart / functional ceramics : PZT, PMN, $BaTiO_3$, ferrites.

Applications: Cutting tools, furnace linings, spark plugs, capacitors, substrates for ICs, bio-implants (hydroxyapatite), sensors and actuators.

1.5 Comparison of Metals, Polymers and Ceramics

Property	Metals	Polymers	Ceramics
Bonding	Metallic	Covalent + secondary	Ionic / covalent
Density	High	Low	Medium
Strength	High, tough	Low	High in compression, brittle
Ductility	Ductile	Flexible / viscoelastic	Brittle
Electrical conductivity	Good conductors	Insulators	Insulators (some semi-conducting)
Thermal stability	Medium - High	Low	Very high

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Corrosion resistance	Poor (rust)	Good	Excellent
Processing	Casting, forging, machining	Moulding, extrusion	Sintering, pressing

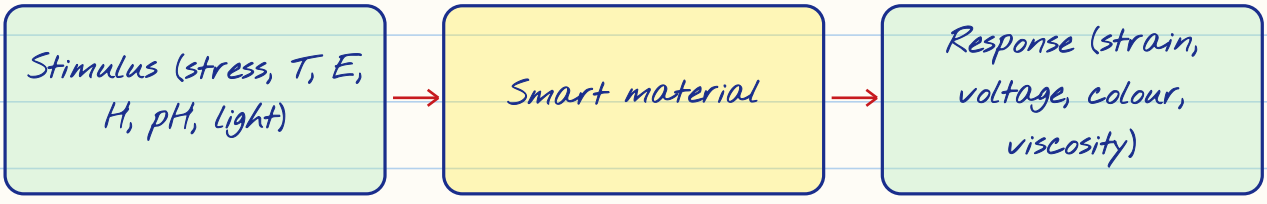
IMPORTANT: Composites combine two materials (matrix + reinforcement) to get properties that none of them has alone, e.g. carbon-fibre epoxy. Many smart structures are made by embedding smart materials inside composites.

1.6 Introduction to Smart Materials

Definition: Smart materials (intelligent / responsive materials) are materials that have one or more properties that can be significantly changed in a controlled fashion by an external stimulus such as stress, temperature, moisture, pH, electric field or magnetic field.

- The response is reversible, repeatable and immediate, and the material "knows" the change in its environment (sensing) and reacts (actuation).
- The term "smart material" was coined in the late 1980s (Japan / USA, 1989 – the "intelligent materials" concept).
- A conventional material only carries load; a smart material senses + decides + responds.

Stimulus -> Smart material -> Response



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Basic features (Cross's 1990s list)

- Sensing - detects the change in the environment.
- Actuation - responds by changing shape, stiffness, position or frequency.
- Immediacy - fast response time.
- Transiency - responds to the active environment only.
- Self-actuation - response comes from within the material.
- Selectivity - response is discrete and predictable.
- Directionality - response is in a definite direction.
- Reversibility - material comes back to the original state when the stimulus is removed.
- Repeatability and long fatigue life.

Smart vs conventional materials

Conventional material	Smart material
Passive - fixed properties	Active - properties change with stimulus
Only sustains the load	Senses and responds to load / environment
Needs separate sensors and actuators	Sensing / actuation built into the material itself
Cannot self-repair or adapt	Can adapt, self-heal or self-sense
Cheap, well established	Costly, new technology, still developing

Terms to remember

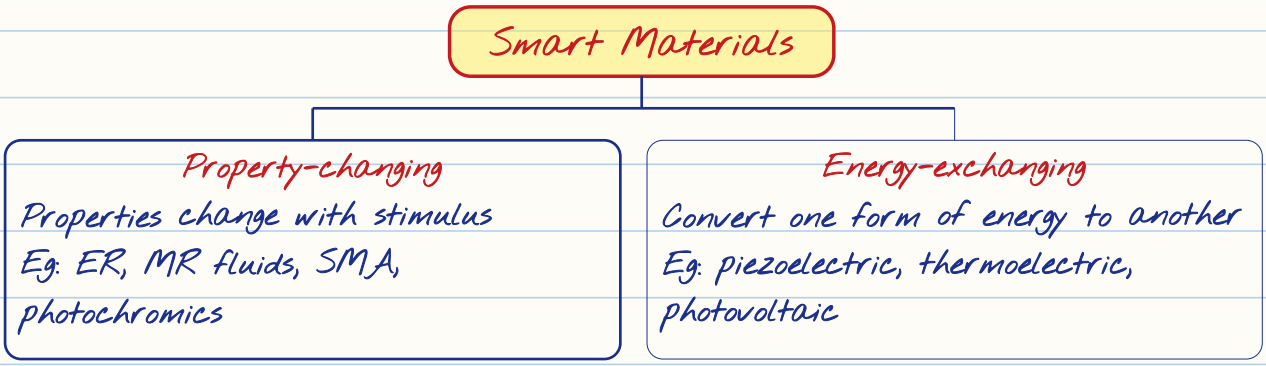
- Smart structure = structure with sensors + actuators + control unit integrated, that can respond to a change in its surroundings.
- Adaptive structure = a structure that can change its geometry / properties to adapt to a load condition.
- Intelligent material = smart material that also has a learning / decision-making ability (microprocessor / feedback control).

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- Multifunctional material = one that performs more than one task (load carrying + sensing).

1.7 Classification of Smart Materials

(a) Based on the type of response



- Type 1 - property changing materials: colour, viscosity, shape, stiffness changes (chromic materials, ER/MR fluids, SMA).
- Type 2 - energy exchanging materials: input energy of one kind → output of another kind (piezo: mechanical ↔ electrical).

(b) Based on function (level of smartness)

- Passive smart materials - can only sense a stimulus, they cannot respond (optical fibre, piezo sensor, strain gauge).
- Active smart materials - sense and actuate (SMA, piezoelectric actuator, magnetostrictive, ER / MR fluids).
- Very smart (intelligent) materials - sense + actuate + adapt their response to changing conditions (needs controller).
- Intelligent materials - can learn and make decisions, like a living organism (bio-inspired).

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(c) Based on material / principle (common list)

Smart material	Stimulus ⟶ Response
Piezoelectric	Mechanical stress ⟷ electric voltage
Electrostrictive	Electric field ⟶ strain (quadratic)
Magnetostrictive	Magnetic field ⟷ strain
Magnetoelectric	Magnetic field ⟷ electric polarisation
Shape memory alloys / polymers	Temperature / stress ⟶ shape recovery
Electro- and Magneto-rheological fluids	Electric / magnetic field ⟶ viscosity change
Thermochromic	Temperature ⟶ colour change
Photochromic	Light ⟶ colour change
Electrochromic	Voltage ⟶ colour / opacity change
pH-sensitive polymers / hydrogels	pH ⟶ volume swelling
Thermoelectric	Temperature difference ⟷ voltage
Self-healing materials	Damage (crack) ⟶ automatic repair
Optical fibres	Strain / temperature ⟶ change in light signal

Other types:

- Chromic materials, electroactive polymers (EAP), ionic polymer-metal composites (IPMC), hydrogels, magnetic shape memory alloys, ferrofluids, self-healing polymers, bio-materials.

1.8 Components of a Smart System

Definition: A **smart (intelligent) system** is a system that can sense the environment, process the information and respond (act) automatically through a **closed feedback loop**.

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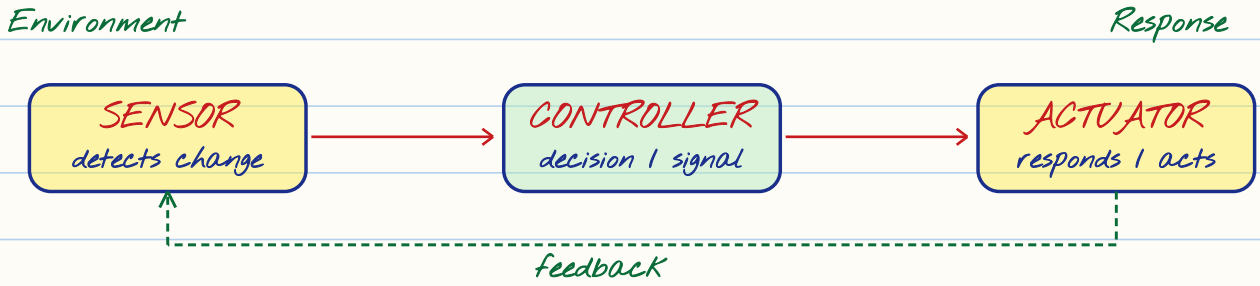


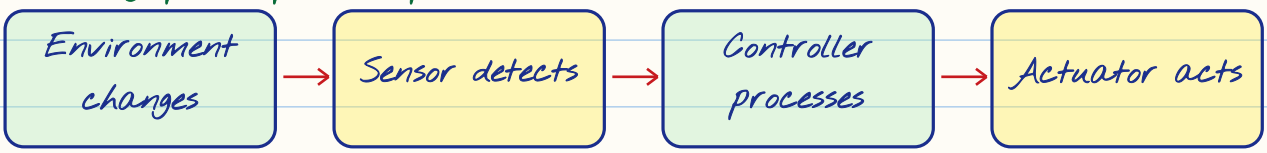
Fig: Components of a smart system

Main components

- 1. Sensors - detect the change in the environment (strain, temperature, pressure, vibration, light, pH) and convert it into an electrical signal.
  - Eg: piezoelectric sensor, fibre-optic sensor, strain gauge, thermocouple.
- 2. Signal conditioning / data acquisition - amplify, filter and digitise the weak sensor signal (amplifier, ADC).
- 3. Controller / processor - the "brain": compares the measured value with the set value (control algorithm such as PID, fuzzy, neural network) and decides the action.
- 4. Actuators - convert the electrical command into mechanical action (displacement, force) to change the system's behaviour.
  - Eg: PZT stack, SMA wire, MR damper, magnetostrictive rod.
- 5. Power source / power amplifier - supplies energy to sensors, controller and actuators (high voltage amplifier for piezo, current driver for SMA).
- 6. Host structure - the main body (beam, wing, plate, bridge) in which the smart elements are embedded or bonded.
- 7. Feedback loop - information about the result is fed back to the controller so that the error is reduced (closed-loop control).

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Working principle (steps)



- Result: structure adapts itself, e.g. a vibrating beam is damped, a wing changes shape, a window darkens.

Types of control

- Open loop - no feedback; action is fixed. Closed loop - uses feedback; accurate and adaptive.

IMPORTANT: Sensors and actuators can be the same smart material : e.g. piezoelectric ceramic works as a sensor (direct effect) and as an actuator (converse effect). Such a material is called "self-sensing".

Examples of smart systems

- Active vibration control of a beam: piezo patches (sensor + actuator) + controller.
- Anti-lock braking system (ABS), airbag system, adaptive headlights in cars.
- Smart window: electrochromic glass that dims with voltage.
- Structural health monitoring of a bridge: fibre-optic sensors + data logger + alarm.

1.9 Applications of Smart Materials

Field	Applications
Aerospace	Active vibration and flutter control, morphing wings, shape memory actuators for flaps, health monitoring of airframe
Civil engineering	Structural health monitoring of bridges and dams, MR dampers for earthquake protection, self-healing concrete

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Automotive	Piezo fuel injectors, airbag sensors, ABS, MR suspension and clutches, electrochromic rear-view mirrors
Biomedical	Stents (NiTi), orthodontic wires, drug delivery hydrogels, ultrasound transducers, artificial muscles, catheters
Consumer electronics	Piezo buzzers, touch screens, camera auto-focus, haptic feedback, smart glasses (photochromic lenses)
Robotics and MEMS	Micro-grippers, micro-pumps, artificial muscles, micro-positioning stages
Textiles	Thermo-regulating fabrics, wearable sensors, smart clothing
Energy	Piezo energy harvesting, thermoelectric generators, solar cells, smart windows
Defence and marine	Sonar transducers (Terfenol-D, PZT), noise reduction in submarines, camouflage
Industrial	Vibration dampers, precision machining, ultrasonic cleaning, MR clutches, pressure and gas sensors

Advantages of smart materials

- Light weight, compact, fast response, low power, high reliability, multi-functional, integration with electronics.

Limitations

- High cost, limited actuation strain (piezo ~0.1%), need of high voltage / power amplifier, hysteresis and non-linearity, ageing, temperature dependence, complex control.

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1.10 Quick Revision / Important Questions

- 1. Compare the characteristics of metals, polymers and ceramics with suitable examples.
- 2. Define a smart material. Explain its basic features with examples.
- 3. How are smart materials classified? Explain with a neat diagram.
- 4. Draw the block diagram of a smart system and explain each component.
- 5. Write short notes on passive, active and very smart materials.
- 6. Discuss any five applications of smart materials in engineering.

IMPORTANT: Exam tip: always draw the block diagram (Sensor → Controller → Actuator with feedback) and add one example for each component. It fetches full marks.

One-line recall

- Smart material = senses + responds reversibly to a stimulus.
- Piezo → force ↔ voltage; SMA → temperature → shape; MRIER → field → viscosity.
- Passive smart = sensor only; Active smart = sensor + actuator; Very smart = + adapts.